

citemsp: Citation Locators for L^AT_EX

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Abstract

We present `citemsp`, a L^AT_EX package that adds per-key locators to numeric citation labels, rendered as compact superscript/subscript pairs¹. Working on top of either `biblatex` or `natbib`, the package provides a slash-delimited syntax with ten built-in prefix types — covering sections, equations, theorems, figures, and more — and a user-extensible registry for domain-specific locators.

1 Syntax and Usage

The `\citemsp` command accepts a comma-separated list of citation keys with optional locators separated by forward slashes. By default, the first locator slot renders as § (section) and the second as ¶ (paragraph). A single-letter prefix changes the locator type. With no locators, `\citemsp` behaves identically to `\cite`, including range compression when using `numeric-comp` or `sort&compress`.

<code>\citemsp{key}</code>	→	$[1]$
<code>\citemsp{key/section}</code>	→	$[1]^{§6.2}$
<code>\citemsp{key/section/paragraph}</code>	→	$[1]^{§6.2}_{¶3}$
<code>\citemsp{key//paragraph}</code>	→	$[1]_{¶3}$
<code>\citemsp{key/section/e4.3}</code>	→	$[1]^{§6.2}_{eq.4.3}$

Currently there exist ten built-in prefixes, which are listed in Table 1. Each prefix can appear in either the superscript or subscript slot. Users can register additional prefixes with `\citemspprefix{letter}{label}`.

2 Examples

¹The package is available at <https://github.com/ApostolosTsabodimos/citemsp-Latex-package> and on CTAN.

Prefix	Locator	Render	In super slot	In sub slot	Both slots
A	Appendix	App.	$[1^{\text{App.2}}]$	$[1^{\text{App.2}}]$	$[1^{\text{§3.2}}_{\text{App.2}}]$
C	Corollary	Cor.	$[1^{\text{Cor.3}}]$	$[1^{\text{Cor.3}}]$	$[1^{\text{§4.1}}_{\text{Cor.3}}]$
d	Definition	$\triangleq$	$[1^{\text{§5}}]$	$[1^{\text{§5}}]$	$[1^{\text{§3.2}}_{\text{§5}}]$
e	Equation	eq.	$[1^{\text{eq.4.3}}]$	$[1^{\text{eq.4.3}}]$	$[1^{\text{§6.2}}_{\text{eq.4.3}}]$
f	Figure	fig.	$[1^{\text{fig.3}}]$	$[1^{\text{fig.3}}]$	$[1^{\text{§6.1}}_{\text{fig.3}}]$
L	Lemma	Lem.	$[1^{\text{Lem.1}}]$	$[1^{\text{Lem.1}}]$	$[1^{\text{§3.2}}_{\text{Lem.1}}]$
n	Footnote	†	$[2^{17}]$	$[2^{17}]$	$[2^{\text{§1}}_{17}]$
p	Page	pp.	$[1^{\text{pp.42}}]$	$[1^{\text{pp.42}}]$	$[1^{\text{§3.2}}_{\text{pp.42}}]$
t	Table	$\boxplus$	$[3^{\boxplus2}]$	$[3^{\boxplus2}]$	$[3^{\text{§8.4}}_{\boxplus2}]$
T	Theorem	Th.	$[1^{\text{Th.4.2}}]$	$[1^{\text{Th.4.2}}]$	$[1^{\text{§6.1}}_{\text{Th.4.2}}]$

Table 1: Built-in prefix registry.

2.1 Basic Usage

The Schwarzschild metric~\citemsp{wald1984/6.1/2} describes spherically symmetric vacuum solutions.

Output: The Schwarzschild metric $[1^{\text{§6.1}}_{\text{§2}}]$ describes spherically symmetric vacuum solutions.

Multiple sources with independent locators:

Black hole thermodynamics~\citemsp{hawking1974/1/2, wald1984/12.5/1} establishes a deep connection between gravity and quantum mechanics.

Output: Black hole thermodynamics $[2^{\text{§1}}_{\text{§2}}, 1^{\text{§12.5}}_{\text{§1}}]$ establishes a deep connection between gravity and quantum mechanics.

Locator entries can be freely mixed:

General covariance~\citemsp{einstein1915, wald1984/4, hawking1974/1/2} forms the foundation of general relativity.

Output: General covariance $[4, 1^{\text{§4}}, 2^{\text{§1}}_{\text{§2}}]$ forms the foundation of general relativity.

2.2 Similarity to \cite

With no locators, \citemsp passes keys through the backend’s native \cite, producing identical output:

\cite{wald1984} $\longrightarrow [1]$ \citemsp{wald1984} $\longrightarrow [1]$

All plain (5 consecutive): $[1\text{--}5]$

All plain (non-consecutive): $[1, 4\text{--}8]$

Range, then locator: $[1, 4, 5, 2^{\text{§3.2}}]$

Locator, then range: $[4^{\text{§3.2}}, 1\text{--}3, 5]$

Range, locator, range: $[1, 4, 5, 2^{\text{eq.2.1}}, 3, 6, 7]$

Adjacent locators only: $[1^{\text{§3.1}}, 2^{\text{eq.2.1}}, 3^{\text{pp.42}}]$

2.3 Mixed Types and Custom Prefixes

All built-in locator types can be freely mixed in a single call:

```
See~\citemsp{wald1984/T6.2, einstein1915/3.2/d5,  
mtw1973/p520, hawking1974/1/e4.3} for further details.
```

Output: See [$1^{\text{Th.6.2}}$, $4_{\text{d5}}^{\S 3.2}$, $3^{\text{pp.520}}$, $2_{\text{eq.4.3}}^{\S 1}$] for further details.

Users can register domain-specific prefixes with `\citemspprefix`. For example, a cosmology paper might define prefixes for window functions and sky patches:

```
\citemspprefix{w}{w.}  
\citemspprefix{S}{\ensuremath{\Omega}}  
The top-hat window~\citemsp{wald1984/4.2/w3} smooths  
the power spectrum over sky patch~\citemsp{mtw1973/S12}.
```

Output: The top-hat window [$1_{\text{w.3}}^{\S 4.2}$] smooths the power spectrum over sky patch [$3^{\Omega 12}$].

2.4 Paragraph example

The following paragraph illustrates how `\citemsp` reads in running text, mixing plain citations, section locators, and prefix types:

```
The Schwarzschild solution~\citemsp{schwarzschild1916/1/3}  
was found shortly after the field  
equations~\citemsp{einstein1915} were published. A modern  
treatment can be found in standard  
textbooks~\citemsp{wald1984/6.1/2, mtw1973/p520}, where the  
derivation proceeds via the Birkhoff  
theorem~\citemsp{wald1984/T6.2}. The singularity  
theorems~\citemsp{penrose1965, hawking1974/1/e4.3} later  
showed that gravitational collapse is generic. For further  
context see~\citemsp{kerr1963, chandrasekhar1931}.
```

Output: The Schwarzschild solution [$5_{\text{1}}^{\S 1}$] was found shortly after the field equations [4] were published. A modern treatment can be found in standard textbooks [$1_{\text{2}}^{\S 6.1}$, $3^{\text{pp.520}}$], where the derivation proceeds via the Birkhoff theorem [$1^{\text{Th.6.2}}$]. The singularity theorems [$6, 2_{\text{eq.4.3}}^{\S 1}$] later showed that gravitational collapse is generic. For further context see [$7, 8$].

3 Loading and Configuration

With biblatex

```
\usepackage[style=numeric-comp, backend=biber]{biblatex}  
\addbibresource{refs.bib}  
\usepackage{citemsp}
```

Use `numeric-comp` (or `numeric`) as the style. Range compression of consecutive plain entries requires `numeric-comp`.

Configuration

The locator glyph size is controlled by `\citemsspscale` (default 0.35):

Default (0.35): $[1^{6.1}_{eq.4.3}]$ Smaller (0.25): $[1^{6.1}_{eq.4.3}]$ Larger (0.50): $[1^{6.1}_{eq.4.3}]$

To adjust the locator glyph size, redefine `\citemsspscale` after loading:

```
\usepackage{citemsp}  
\renewcommand{\citemsspscale}{0.4} % default is 0.35
```

4 Relation to Standard Locators

The standard `biblatex` provides locators via its postnote mechanism, the `\citemsp` package suggests a more compact mechanism:

```
% Standard biblatex:  
\autocites[$6.1, ¶2]{wald1984}[pp.\,520]{mtw1973}[eq.\,4.3]{hawking1974}  
  
% citemsp equivalent:  
\citemsp{wald1984/6.1/2, mtw1973/p520, hawking1974//e4.3}
```

Three advantages distinguish `\citemsp`: (1) a compact, uniform syntax that stays readable as entries accumulate; (2) typed locator prefixes that visually distinguish sections, equations, pages, theorems, and other reference types; and (3) superscript/subscript rendering that keeps locator information close to each citation number rather than appended as flat text.

5 Conventions

- **Section numbering.** Use the source’s native section numbering format (e.g., 6.1 for “Section 6.1”).
- **Paragraph counting.** Paragraphs are counted manually from the start of the section. This is inherently edition-dependent and best suited for stable references such as textbooks.
- **Custom prefixes.** Register new locator types with `\citemspprefix{letter}{label}`. The letter must be a single character not already in use. Custom and built-in prefixes can be freely mixed in the same `\citemsp` call.

References

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- [1] Robert M. Wald. *General Relativity*. University of Chicago Press, 1984.
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 - [3] Charles W. Misner, Kip S. Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman, 1973.
 - [4] Albert Einstein. “Die Feldgleichungen der Gravitation”. In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1915), pp. 844–847.

- [5] Karl Schwarzschild. “Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie”. In: *Sitzungsberichte der Preussischen Akademie der Wissenschaften* (1916), pp. 189–196.
- [6] Roger Penrose. “Gravitational Collapse and Space-Time Singularities”. In: *Physical Review Letters* 14 (1965), pp. 57–59.
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- [8] Subrahmanyan Chandrasekhar. “The Maximum Mass of Ideal White Dwarfs”. In: *The Astrophysical Journal* 74 (1931), pp. 81–82.